

LM 1: Div & Repurchase · 利润分配问题.

Dividend

- 1 4 Types: Cash / Stock / Split / Reverse. $\Rightarrow$ 影响
- 2 5 Theory
发不发
 - ① does not matter (假设 perfect market)
 - ② bird in hand: 在手确定性, 应发.
 - ③ tax aversion: div tax 高于 capital gain tax $\rightarrow$ 不发或少发.
 - ④ Signal: 发股利是利好信号, 股价 $\uparrow \rightarrow$ 发.
 - ⑤ agency cost: 发解决代理人风险 (ex. overinvestment)
- 3 6 Factors
多发或少发.
 - Investment Opportunity Div $\downarrow$
 - Expected Volatility Div $\downarrow$
 - Financial Flexibility (Repurchase 比发股利更灵活) Div $\downarrow$
 - Flotation Cost 发行成本 Div $\downarrow$
 - Contractual & Legal Restrictions
 - Tax Consideration
 - 3种 税制
 - ① Double Taxation 双重征税制. 税负重.
 - ② Split tax rate system 分拆征税制 (Div tax 低, P/E tax 高)
 - ③ Tax imputation system 不承担企业所得税.
 - 计算 effective tax rate
- 4 Payout Policy 发多少
 - ① Constant Div Payout Ratio Policy
 - ② Stable Div Policy $\checkmark$ $\text{The expected increase in dividends} = (\text{Expected earnings} \times \text{target payout ratio} - \text{previous dividend}) \times \text{adjustment factor} \times \frac{1}{\text{year}}$

Repurchase

- 1 4 Methods
 - open market
 - tender offer 通常比 market price 高.
 - dutch auction
 - negotiation (与大股东谈).
- 2 对财务状况影响
 - D/E (Leverage) $\uparrow$
 - 回购资金成本与 earnings yield 相比. 买得划算. EPS $\uparrow$
 - BVPS 每股净资产. 与 market price 相比. $BVPS > \text{market price}$. BVPS $\uparrow$
- 3 优势
 - tax rate 低
 - signal
 - flexibility
 - # $\downarrow$ offset dilution.

Div Safety

- 指标
- dividend coverage ratio = net income / dividend paid 越大越好, 通常 > 1
 - FCFE coverage ratio = FCFE / (dividends + share repurchases)

LM2 ESG

Ownership Structure

- 1 类型 } 持股比例是否集中: Dispersed & Concentrated
持股方式: Horizontal (交叉持股) & Vertical
- 2 Conflicts & Problems } Principal Agent Problem - 股权、投票权均分散
Principal Principal Problem { 股权、投票权均集中 (小股东利益无保障)
股权分散, 投票权集中
Voting Cap Restrictions - 股权集中, 投票权分散
- 3 Dual-Class Shares
- 4 Effects on Corporate Governance

董事会独立性/董监结构/投票权 **cumulative** 保护中小股东/ 准则/ Stewardship Codes 尽责管理准则 investor 参与

Evaluating Corporate Governance

- 1 Shareholder Activism / Shareholder Engagement
- 2 BOD 质量
 - independence
 - committees 委员会设置
 - skill & experience
 - composition 多样化
- 3 Executive Compensation
 - transparency
 - linkage
 - clawback policy

ESG

- ① Identify
- ① primary challenge: inconsistent & voluntary
评估标准不一致 自愿披露
 - ② materiality (分析重要的) & horizon (影响长期 > 短期)
 - ③ 获取信息
 - Proprietary Method - analyst 自己
 - ESG Data Providers - 外部机构
 - Non-profit - 非盈利
- ② Evaluating
- Equity: ESG对股票估值的影响. 估值
 - Fixed Income: 更关注风险, spread ↑ Credit Spread

LM3 Cost of Capital = Required Return

$$r_d = r_f + \text{Credit spread}$$

$$r_e = r_f + \text{ERP} + \text{IRP}$$

Factor

Top-Down, External	Bottom-Up, Company Specific
Capital availability	Revenue, earnings, and cash flow volatility
Market conditions	Asset nature and liquidity
Legal and regulatory considerations/country risk	Financial strength, profitability, and leverage
Tax jurisdiction	Security features

Cost of Debt 计算方法

- ① $r_f + RP + RP + \dots$ 加各类风险溢价。就是债券要回报率。
- ② $PV, FV, PMT(\text{coupon}), N \Rightarrow I/Y$ (年化) $\Rightarrow YTM$ 前提是不含权且活跃交易
- ③ Matrix Pricing 用其他债券 YTM 合成
- ④ 若外国: + Country Risk 范围大于 Sovereign Risk (国家违约)

优先股: 本质是永续年金
 $P_0 = \frac{C}{r} \rightarrow r = \frac{C}{P_0}$

ERP

- ① r_e 计算方法:
 - ① DDM: $P_0 = \frac{D_1}{r_g} \rightarrow r_e = \frac{D_1}{P_0} + g$
 - ② Bond Yield plus RP: $r_e = \text{Yield} + RP$ before tax
 - ③ Risk Based Model: $r_e = r_f + RP + RP + \dots$
- ② ERP
 - Historical Approach 历史平均 10% 平均 the best!
 - Index / Period / Mean / Proxy for R_f
 - Forward-looking
 - Survey
 - DDM
 - Macroeconomic Model: GK Model
 - Grinold-Kroner Model (ERP)

$$\text{ERP} = [DY + \Delta(P/E) + i + g + \Delta S] - E(r_f)$$
 Div + Capital Gain

CAPM
 PFM
 PSM

r_e

- ① CAPM: $r_e = r_f + \beta \cdot \text{ERP}$ 只考虑系统风险, 未考虑 individual RP

- ② Multifactor Model
 - FFM 3
 - PSM 3+ Liquidity

$$\begin{aligned} \text{Required Return} &= R_F + \beta_{\text{mkt}}^{\text{mkt}} \times (R_{\text{mkt}} - R_F) \\ \text{Small/large cap} &\rightarrow +\beta_{\text{size}}^{\text{size}} \times (R_{\text{small}} - R_{\text{big}}) \\ \text{High/low book-to-market} &\rightarrow +\beta_{\text{value}}^{\text{value}} \times (R_{\text{HBM}} - R_{\text{LBM}}) \\ \text{Low/high liquidity} &\rightarrow +\beta_{\text{liq}}^{\text{liq}} \times (R_{\text{LL}} - R_{\text{HL}}) \end{aligned} \quad \left. \begin{array}{l} \text{FFM} \\ \text{PSM} \end{array} \right\}$$

$\beta_{\text{size}} > 0$, small cap
 $\beta_{\text{value}} > 0$, value-oriented
 $\beta_{\text{liq}} > 0$, low-liquidity

- ③ Non-public β : pure-play 去/加杠杆

$$\begin{aligned} \beta_u &= \left[\frac{1}{1 + (1-t) \frac{D}{E}} \right] \cdot \beta_E \quad \text{上市公司} \\ \beta_E &= \left[1 + (1-t) \frac{D}{E} \right] \cdot \beta_u \quad \text{非上市公司} \end{aligned}$$

- ④ Private Company 其 Risk size risk / industry risk / specific-company risk

$$\begin{aligned} \text{Expanded CAPM: } r_e &= r_f + \beta \cdot \text{ERP} + \text{SP} + \text{IP} + \text{SCR} \\ \text{Build-up } (\beta=1) &\text{ 多因素模型中默认非上市公司 } \beta=1 \end{aligned}$$

- ⑤ International
 - CRP: $\text{CRP} = \text{Sovereign yield spread (两国国债之差)} \times \frac{G_{\text{equity}}}{G_{\text{bond}}}$
 - ICAPM: $\text{ERP} = \text{ERP for a developed country} + \lambda \cdot \text{CRP}$ 与发达国家 ERP 合并

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LM4. Corporate Restructuring

Concepts & Types

	Investment Actions	Divestment Actions	Restructuring Actions
Issuer-specific motivations	- Realize synergies - Increase growth - Improve capabilities or secure resources - Opportunity to acquire an undervalued target	- Focus operations and business lines - Valuation - Liquidity needs - Regulatory requirements	- Improve returns on capital - Financial challenges, including bankruptcy and liquidation
Types	Equity Investment Joint Venture Acquisition	Sale Spin off 分拆 (股东保持不变)	Costs Balance sheet Reorganization

sale leaseback
div recapitalization

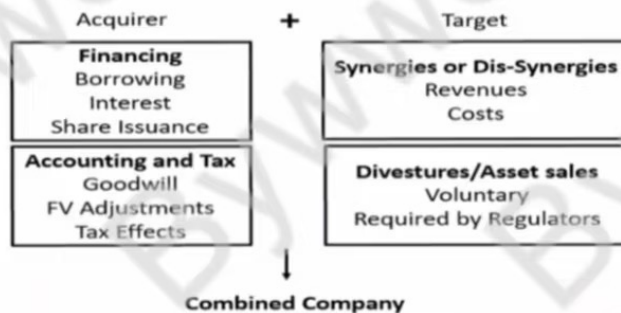
Initial Evaluation

What / why / material? / time (取决于项目复杂程度及审批时间)

Preliminary Valuation
 用作估值系数: } Comparable Company
 Comparable Transaction 无溢价 + takeover premium

Modelling & Valuation

1 Pro Forma P/S 模拟调整收购后财报



2 Pro Forma WACC 用以折现:

Evaluating 收购后回头看效果

- Investment: synergy < scale 规模效应
 scope 范围效应 经营范围扩大, 降低成本
- Divestment: conglomerate discount 集团效应导致估值低.
 根据估值决定 sale or spin off
- Restructuring: opportunistic improvement (主动), forced improvement